Causal Inference

Overview

- what is Causality, and how is it different from correlation?
- foundational concepts of causal inference
- avoiding common pitfalls

This Week's Goals

- distinguish between descriptive, predictive, and causal analyses
- the notion of confounding
- the role of un-observables in causal attribution
- the notion of balance
- the value of randomization in causal attribution
- packages needed
    1. R: tidyverse
    2. R: gridExtra

Readings

Causal Inference: Chapter 1

Scott Cunningham

ISBN 9780300251685

https://mixtape.scunning.com/01-introduction

Chapter 1 Summary	this chapter argues that **causal inference** is **necessary** because real-world **data** often **contain** strategic human **choices**, hidden **confounders**, and **misleading correlations**, so researchers need **theory**, **assumptions**, and **careful** research **designs** to **estimate** **causal effects** credibly

Introduction	• this chapter explains **why causal inference matters**
	    ○ **we** often **want** to **know** whether X **causes** Y
	    ○ **most data** only **show** that X and Y **move together**
	    ○ **correlation** may (or may not) **suggest causality**
	• causal inference gives **researchers tools** for **estimating causal effects** when **randomized experiments** are **not available**

Core Concepts
Glossary
	causal inference
	    estimating whether one thing **causes another** thing
	correlation
	    two variables **move together**
	    correlation alone does **not** prove **causation**
	causation
	    one **variable directly affects** another **variable**
	observational data
	    data where the **researcher observes** what happened
	    does **not control treatment** assignment
	experimental data
	    data where the **researcher** has some **control** over **treatment assignment**
	endogeneity
	    a **variable** is **connected** to **hidden factors** that also **affect** the **outcome**
	    this **makes** causal **interpretation difficult**
	exogeneity
	    a variable **changes independently** of **hidden factors** that **affect** the **outcome**
	    this **makes** causal **interpretation** more **credible**
	identification

	process of **showing** that a **research design** can **recover** the **causal effect** of interest **ceteris paribus** "All else equal." a **causal claim** usually **asks** what **happens** when one thing **changes** while relevant **other** things **stay fixed** **price elasticity of demand** **measure** of how much **quantity demanded changes** when **price changes**
What is causal inference?	**causal inference** is the **process** of using • **theory** • institutional **knowledge** • **research** design • **data** • statistical **methods** to **estimate** the **effect** of one thing on another example • does **education** increase **income**? • does a **policy** reduce **crime**? • does **price** affect **demand**? • does a **medical treatment** improve **health**? the author defines **causal inference** as • using **theory** and **deep contextual knowledge** to **estimate** how **events** or **choices affect** an **outcome**

<table>
<tr>
<td>Why this book exists</td>
<td>
the author argues that many causal inference books are valuable but incomplete for applied learners

some books emphasize

theory
potential outcomes
DAGs
econometrics

the author wanted a book that combines

readable explanation
research design
programming examples
real data
replication
and practical implementation

the book is written for

practitioners
social scientists
industry
policymakers
students
usable tools

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<td>Correlation is not Causation</td>
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common sense tells us

correlation does not mean causation
two things can move together without one causing the other

example

rooster crows → sun rises
rooster did not cause sunrise!

conclusion
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	• simple **correlation** does **not prove** causal **relationships**

No Correlation does not mean No Causation	sometimes **causality exists** even when **correlation** is **hard** to **see**
	
	example
	• imagine a **sailor crossing** a **lake**
	  ○ **wind pushes** the **boat**
	  ○ **sailor moves** the **rudder** to **cancel** the **wind**
	  ○ the **boat travels** in a **straight** line
	• an **observer** might **think**
	  ○ *"the rudder has **no effect**"*
	• that would be **wrong**
	  ○ the **rudder** is **working**
	  ○ its **effect** is **hidden** because the **sailor adjusts** it in **response** to the **wind**
	• **causal** effect **exists**, but the **observed correlation** is **canceled out**

Why Randomization Matters	the **sailboat example** shows why **randomization** is **powerful**
	• if the sailor **randomly moved** the rudder
	  ○ **observer** would **see** the **boat zigzag**
	  ○ **relationship between** rudder **movement** and boat **direction** would become **visible**
	• key **idea**
	  ○ **randomness** helps **reveal** causal **effects**
	• when **people choose** actions **strategically**
	  ○ the **data** can **hide** causal **relationships**
	• when **treatment** is **randomized**

	◦ the **researcher** has a better **chance** of **isolating** the **effect**
Optimization Makes Causal Inference Difficult	• **people** make **choices** based on ◦ **goals** ◦ constraints ◦ **incentives** ◦ expectations ◦ private **information** ◦ and **predicted outcomes** • because of this ◦ many **choices** are **endogenous** • that **means** ◦ the **choice** is **connected** to the **outcome** we are trying to **study** • this **creates** a major **problem** ◦ if **people choose treatment** based on **expected outcomes** ▪ treatment is **not independent** of the **outcome** • meaning ◦ a simple **correlation** between **choice** and **outcome cannot** be **assumed** as **causal**
Experimental Data vs. Observational Data	example ▪ a **randomized** medical **experiment** ▪ a **field experiment** ▪ a **policy experiment** with **random assignment** this is often **stronger** for **causal inference** because ▪ the **researcher** has more **control** over **how** treatment is **assigned**

	in **observational** data
	▪ the **researcher**
	    o usually does **not control** what **happens**
	    o **observes choices** and **outcomes after** they occur
	▪ examples
	    o **surveys**
	    o administrative **records**
	    o **business** data
	    o **web-scraped** data
	    o **historical** data
	▪ most **social science** and **business data** are
	    o **observational**
	▪ **observational data** are
	    o **harder** to use for **causal claims** because
	    o **choices** are often **endogenous**

Causal Inference Needs Assumptions	▪ **causal inference**
	    o **not** just **mechanical statistics**
	▪ **good** causal **research requires**
	    o **clear assumptions**
	    o **theory**
	    o **local knowledge**
	    o **research design**
	    o **honesty** about **uncertainty**
	    o **process-oriented scientific attitude**
	
	the **author warns** against
	▪ trying to **force results** to **match** what the **researcher wants**
	▪ causal **methods** are only **credible** when

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- the **researcher prioritizes** correct **method** over desired **outcome**

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Example: Price Elasticity of Demand

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- price **elasticity** of **demand** to show why **causal inference** is **hard**
- researchers **want** to **know**
  - if **price changes**, how much does **quantity demanded change**?
- **elasticity formula**

$$\epsilon = \frac{\partial log P}{\partial log Q} \qquad where,$$

$$\boldsymbol{Q} = quantity$$

$$\boldsymbol{P} = price$$

$$\boldsymbol{\epsilon} = \boldsymbol{price\ elasticity}\ of\ demand$$

- this **measures**
  - the percentage **change** in **quantity** associated with a percentage **change** in **price**

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Why price and quantity data are not enough

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- we do **not directly observe** a full **demand curve**
- instead we usually **observe market equilibrium points**
  - **one** price
  - **one** quantity
- those **points** may **come from**
  - **supply** shifts
  - **demand** shifts
- **connecting** observed **price-quantity** points
  - may **not** trace demand curve
- result

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	- a simple **correlation** between **price-quantity** may **not** estimate **demand elasticity** ▪ to estimate **demand elasticity** - price **variation** must be **independent** of ▪ **supply** ▪ **demand**
Causal Inference: Chapter 2 **Scott Cunningham** **ISBN 9780300251685** **https://mixtape.scunning.com/02-probability_and_regression**	
Chapter 2 Summary	Chapter 2 teaches the **probability**, **expectation**, **variance**, **covariance**, **OLS**, **standard error**, and **regression** anatomy **tools** needed for **causal inference**, while repeatedly **emphasizing** that regression only supports causal claims when the key assumption $\boldsymbol{E(u \mid x) = 0}$**is** credible
Introduction	▪ this chapter reviews the probability and regression tools needed for causal inference - regression can estimate useful relationships - causal interpretation depends on assumptions ▪ the most important assumption in the chapter is $$E(u \mid x) = 0$$ ▪ this means - the unobserved causes of y must not systematically change with x ▪ if this fails - regression may still produce a number

	○ that number may not be causal

Why Probability Comes First	causal inference uses statistical models
	▪ those models depend on probability concepts like
	  ○ random processes
	  ○ sample spaces
	  ○ independence
	  ○ conditional probability
	  ○ expected value
	  ○ variance
	  ○ covariance
	  ○ regression
	▪ a random process
	  ○ can be repeated
	  ○ may produce different outcomes each time
	▪ sample space
	  ○ the set of all possible outcomes
	▪ examples
	  ○ discrete
	    ▪ coin flip
	    ▪ die roll
	    ▪ card draw
	  ○ continuous
	    ▪ gas price

Independence	▪ two events are statistically independent if ○ knowing one happened does not change the probability that the other happened $$Pr(A \mid B) = Pr(A)$$

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meaning
the probability of A, given B, is the same as the probability of A alone
example
independent
rolling a 3 on one die does not affect rolling a 5 on another die
not independent
drawing cards without replacement
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<tr><td>Joint Probability</td><td>
joint probability
the probability that two events both happen
for independent events

$$Pr(A, B) = Pr(A)Pr(B)$$

example
if
Cleveland must win 3 games in a row
each game has probability 0.5
then

$$\mathbf{(0.5)^\wedge 3 = 0.125}$$

so, the probability is

$$\mathbf{12.5\%}$$

key idea
when independent events must all occur
multiply probabilities
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Conditional Probability

conditional probability asks:

- what is the probability of one event, given that another event already happened?

- Formula:

$$\boldsymbol{Pr(B \mid A) = Pr(A)Pr(A,B)}$$

- example from the chapter
  - A: Texas makes a great bowl game
  - B: the coach is rehired

$$\boldsymbol{Pr(A) = 0.6 \qquad Pr(B) = 0.8}$$

$$\boldsymbol{Pr(A,B) = 0.5}$$

- then

$$\boldsymbol{Pr(\,B \mid A\,) = \frac{0.6}{0.5} = 0.83}$$

- so
  - if Texas made a great bowl game
  - the probability the coach is rehired is about 83%
  - but:

$$\boldsymbol{Pr(A \mid B) = 0.8\ 0.5\ \ = 0.625}$$

- so
- if the coach was rehired
- the probability Texas made a great bowl game is about 63%

- important

$$\boldsymbol{Pr(\,B \mid A\,) \neq Pr(A \mid B)}$$

Bayes' Rule	lets us update beliefs after receiving new information ▪ basic form $$Pr(A \mid B) = \frac{Pr(B \mid A)Pr(A)}{Pr(B)}$$ ▪ expanded form $$Pr(A \mid B) = \frac{Pr(B \mid A)Pr(A)}{Pr(B \mid A)Pr(A) + Pr(B \mid \sim A)Pr(\sim A)}$$ ▪ meaning $$posterior\ belief\ = \ updated\ belief\ after\ seeing\ evidence$$ Bayes' rule helps us reason from an observed effect back toward possible causes

Monty Hall Example	the Monty Hall problem shows why conditional probability is unintuitive
	
	the problem
	▪ there are 3 doors
	  ○ one has money
	  ○ two have goats
	▪ you choose one door
	▪ Monty opens another door and shows a goat
	  ○ you can stay or switch
	▪ many people think
	  "Now there are two doors, so it must be 50–50."
	▪ that is wrong
	  ○ your original door still has probability $\frac{1}{3}$

	- the other unopened door has probability $\frac{2}{3}$ - you should switch - because Monty's action gives information - he does not open doors randomly - he opens a door he knows has a goat
Summation Notation	- summation symbol $$\Sigma$$ - example $$\sum_{i=1}^{n} x_i = x_1 + x_2 + \cdots + x_n$$ - the sample mean is $$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$ - a key property $$\sum_{i=1}^{n}(x_i - \bar{x}) = 0$$ - this implies - deviations from the mean always sum to zero - this matters because - regression uses deviations from means to estimate slopes

Expected Value	▪ the probability-weighted average of possible outcomes $$E(X) = \sum_{j=1}^{k} x_j f(x_j)$$ ▪ example ○ if X can be: -1 $with\ probability$ 0.3 0 $with\ probability$ 0.3 2 $with\ probability$ 0.4 ○ then $E(X) = (-1)(0.3) + (0)(0.3) + (2)(0.4) = 0.5$ ○ so, the expected value, 0.5 ▪ is a population concept ▪ describes the average value across the full probability distribution
Variance	variance measures spread ▪ population variance $$V(W) = E[(W - E(W))^2]$$ ▪ equivalent formula $$V(W) = E(W\^2) - E(W)\^2$$ ▪ sample variance $$\widehat{S}^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - x)^2$$

Covariance and Correlation

- measures whether two variables move together

$$C(X,Y) = E(X\,Y) - E(X)E(Y)$$

- if covariance is positive
    - X and Y tend to move in the same direction
- if covariance is negative
    - X and Y tend to move in opposite directions

- but covariance is hard to interpret
    - its size depends on the units of the variables

- correlation standardizes covariance

$$Corr(X,Y) = \frac{C(X,Y)}{\sqrt{V(X)V(Y)}}$$

- correlation is always between

$$-1 \text{ } and \text{ } 1$$

important warning

- zero covariance does not always mean two variables are unrelated
- they may have a nonlinear relationship

Population Regression Model

the basic population model is:

$$y = \beta_0 + \beta_1\, x + u, \quad where$$

$$y = outcome$$

$$x = explanatory\ variable$$

$$\beta_0 = intercept$$

$$\beta_1 = slope$$

$$u = error\ term$$

the error term u contains other causes of y that are not included in the model

	▪ example ○ if y is wages and x is schooling ○ then u may include ▪ ability ▪ family background ▪ motivation ▪ school quality ▪ luck ▪ social networks ○ the researcher ▪ observes x and y ▪ does not observe u
Mean Independence and Zero Conditional Mean	▪ mean independence says $$E(u\|x) = E(u)$$ ▪ if we normalize $$E(u) = 0$$ ▪ we get $$E(u\|x) = 0$$ zero conditional mean assumption ▪ meaning ○ the average value of the unobserved error is the same for every value of x ▪ example ○ if studying schooling and wages

	○ this assumption would require average unobserved ability to be the same across schooling levels ○ that is a strong assumption and may fail because people choose schooling partly based on ▪ ability ▪ ambition ▪ family resources ▪ expectations
Ordinary Least Squares	▪ OLS estimates the line $$\mathbf{y} = \hat{\boldsymbol{\beta}}_0 + \hat{\boldsymbol{\beta}}_1\mathbf{x}$$ ▪ slope estimate $$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$ ▪ can also be described $$\hat{\beta}_1 = \frac{\textit{Sample covariance of x and y}}{\textit{Sample variance of x}}$$ ▪ OLS needs variation in x ▪ if everyone has the same value of x, the slope cannot be estimated
Fitted Values and Residuals	▪ a fitted value is the model's prediction $$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$ ▪ a residual is the prediction mistake $$\hat{u}_i = y_i - \hat{y}_i$$

	▪ error term: $\boldsymbol{u_i}$ ○ unobserved ○ part of the population model ○ contains omitted causes of y ▪ residual: $\boldsymbol{\hat{u}_i}$ ○ observed after regression ○ calculated from data ○ equals the model's prediction error the error term and the residual are not the same thing
What OLS Minimizes	▪ OLS chooses the intercept and slope that minimize the sum of squared residuals $$\sum_{i=1}^{n} \hat{u}_i^2$$ it is called least squares because it chooses the line that makes squared prediction errors as small as possible among linear fits
	▪
	▪

Demand Model	possible econometric demand model
	
	$$logQd = \alpha + \delta logP + \gamma X + u, \quad where$$
	$$Q\_d = quantity\ demanded$$
	$$\alpha = intercept$$
	$$\delta = demand\ elasticity$$
	$$P = price$$
	$$X = other\ demand\ factors$$
	$$\gamma = effect\ of\ those\ other\ factors$$
	$$u = error\ term$$
	examples
	- income
	- prices of other goods
	
	- the goal is to estimate $\boldsymbol{\delta}$
	  - to do that
	  - price must vary independently of the error term u
	
	this independence is called: exogeneity
	
	without both the estimate may be misleading

Key Takeaway	- to estimate price elasticity we need
	  - many observations of
	    - price
	    - quantity
	  - price variation
	
	must be independent of unobserved demand factors
	
	this is the same general problem causal inference tries to solve
	
	- we need either
	  - variation in treatment that is as good as random
	  - research design that makes a credible causal comparison possible